

Using LOGIC 2010

The program provides exercises -- and automated correction -- for first order symbolic logic. It is not, and does not aim to be, a self-contained instruction package. It supplements an instructor's lectures and the study of a text by providing a large number of new exercises and by making use of the computer's facility for immediate evaluation, context sensitive hints, automation of advanced techniques, and adaptation to the stage of advancement of the student.

Work done within the program is corrected by the program, erroneous steps are indicated, and several levels of explanations of the error are provided. The program encourages you to move at your own pace, and to explore alternate ways of approaching problems. You may correct and recheck your work before submitting it. Completed work may be printed or submitted over the Internet to a database, which also provides a personal assignment page showing the assignments that your instructor has created and the work that you have completed (click "Assignments" on the Menu).

The formal system of derivation is a natural deduction system in the style of the textbook:

LOGIC: Techniques of Formal Reasoning, Second Edition
by Kalish, Montague and Mar
Published by Harcourt Brace Jovanovich, 1980
[Henceforth, *KM&M*]

However, the program diverges from this textbook in a several respects. The online textbook:

An Introduction to Symbolic Logic
by Terrence Parsons
[Henceforth, *The TerryText*]

is more closely keyed to the program and is currently available to users of the program without charge.

Different modules allow you to pursue work on different kinds of exercises, including, Derivations, Truth Tables, Parsing, Symbolizations, and Invalidity. There is also a module to help you learn to Recognize Rules.

The program runs on both a PC and a MAC. It is available from a Download site, accessible from the Logic 2010 Portal at:

<http://logic2k.humnet.ucla.edu/>

It will create the folder: Logic 2010. The document "Installing, Starting, Initializing, and Backing Up LOGIC 2010" (available on the Download website) provides step by step instructions.

Your work will be stored in the work sub-folder to LOGIC 2010 on your hard disk. Your work files can be backed up over the Internet to the Logic 2010 database. (Click "Backup" on the Menu.) You should back up at the end of every session. If you move from one computer to another, Restore your most recent backup before you begin work.

Only the logic program may write to your work folder. If you edit the files in this folder with another editor, they will be corrupted, and you will have lost your work.

The first time you use the logic program you will select your institution, term, and course. You will then be given the opportunity to *register* yourself as a Logic 2010 user in the database.

As part of the registration process you are required to create a Logic Password. Your logic password, which is stored in encrypted form, is used to ensure that only you can submit work in your name. If you forget your logic password, your Instructor can reset it. However, the reset process is complex, and may take a day or so. So *remember* your password. Also remember that passwords are always case sensitive. If you have not read "Selecting Good Passwords", or something like it, we suggest that you look at <http://www.alw.nih.gov/Security/Docs/passwd.html>.

STARTING THE PROGRAM

To start the program on a PC, go to the LOGIC 2010 folder, and click on the file Logic 2010.exe. You can place a shortcut to the file Logic 2010.exe on your desktop by right-clicking the file, and selecting "Send To/Desktop (create shortcut)". To start the program on a MAC, just click on the application.

When the program itself opens, you should see a gray screen with the phrase "LOGIC 2010: A Workbook" and a several Version numbers written at the top in blue. There will be a column of buttons below that. This is the Menu; it is the center and jumping-off place for all modules. It is also the only place from which you can Back up and Quit the program. From here, you can also view Assignments, Notices (at the instructors option), and the Course Website (if there is one).

If you wish to work in a new location, say, a computer lab, after having completed some work and using Backup, you can install the program in your new location (unless it is already there), start the program, and then Restore your most recent backup to the lab computer. (See below on how to restore.) When you complete your work in the lab, you should again back up your work using Backup, and then quit the program using "Delete Work/Quit" (rather than just "Quit"). This will clean up your Logic work-folder so that the next person will not be inconvenienced. If you are not in a public facility, such as a computer lab, back up, and then just use "Quit".

SUBMITTING WORK

Work can be printed, and given to your instructor as hard copy, or "Submitted" over the Internet to a database of student work. Note that it is always the last "saved" version of a problem that is printed or submitted.

Work submitted to the database is correlated with Assignments constructed by the instructor. The database provides a report to you, organized by assignment, regarding your own work, and a report to your instructor regarding the work submitted by all students. You can access the assignments and a report on your own work through the program Menu, or independently from the Logic 2010 Portal.

When you submit work, you must verify that the correct term, course, and instructor and TA names are displayed. If not, select the correct choices from the drop-down lists.

COMMUNICATIONS

Common to all modules are communications facilities providing Notices (at your instructor's option), a Feedback facility for student responses, and a connection to the Course Website (at your instructor's option) for the Discussion Board (at your instructor's option). Occasional messages from the Instructor are also posted on the Homepage of the course website.

The Notices document contains information about the program. It will appear automatically when you open the program, but only if it contains new information (information you have not already read). Recent and new items will be marked "**NEW**", and will appear at the top. Please read them. Close the document in the usual way, for example by clicking on the

upper right hand “x”.

Feedback comments are sent to your instructor, TA (if any), and the developers of the program. Use the Feedback facility for issues having to do with the program, especially to report bugs. The Feedback facility is accessible from each module.

The **Discussion Board** on the course website is the best venue for comments or questions concerning logic. That way, your fellow students can help answer questions, and can view answers to questions that they too may have found puzzling. The instructor and TAs check the Discussion Board on a regular basis.

For personal matters, write directly to the instructor or your TA.

You should check the **Course Website** on a regular basis. In addition to the Discussion Board, messages about the course, links to helpful documents, distributions of scores on exams, and other useful information may appear there. Classroom work is also often incorporated directly into the program (as “CL” exercises).

USING THE INDIVIDUAL MODULES

Individual modules are selected from the Menu.

Remember, you need to know some logic to get started. The software does not aim to be a self-contained instructional package.

Hover Messages

If you hover the mouse pointer, without moving, above a button, a hover message will appear describing the use of the button or giving an alternative keyboard command.

Selecting a problem to work on

Click the “Select” button at the bottom of the program window. This will bring up a list of exercises to choose from with the preceding choice highlighted. Exercises not yet attempted appear in BLACK. Exercises done correctly appear in GREEN. Exercises done incorrectly (or incompletely) appear in RED. To select an exercise to work on, highlight it, and press Enter, or highlight it, and click OK, or simply double-click the exercise. You can jump directly to an exercise by typing in its number. You can also select the exercise immediately following the one you are currently working on by right-clicking the Select button. (If the window does not open to the problem you last selected, quickly press UpArrow then DownArrow.)

Creating your own ‘User’ problem

Rather than working on the preset exercises, you may wish to work on a problem of your own, or on one that your instructor has proposed but that has not been programmed into the Exercises list. Such problems are called “User” problems. To create a new User problem, simply click the “User” button, and enter the problem in the User Problem box. You may either type in the problem using the Keyboard Shortcuts (see below) or use the mouse by right-clicking anywhere in the User Problem box to invoke a Toolbox of buttons for entering symbols and performing other relevant operations.

In creating a new problem involving an argument, remember that each premise, except the last, must be followed by a period, and that the “therefore” symbol, which appears on a button in the Toolbox, must appear between the last premise and the conclusion.

User problems are saved to the bottom of the list of exercises. When you save a User problem, you will be prompted to enter a name. The prefix “User” will be suggested, but any name not already in use will work.

Starting work in a module

A Toolbox, invoked by a Right-Click of the Mouse, is common to all modules. (The actual contents of the Toolbox vary by module and Chapter.) It can be used to enter the logical symbols. The hover message associated with each symbol button in the Toolbox displays keystrokes that can also be used to enter that symbol.

Each module contains its own Help document (click “Help” on the button bar), which begins with a paragraph or so headed “Quick Start” to get you started using the module. The Help documents provide help in *learning to use the program*, not in learning logic.

In the case of the Derivation module, which contains many special tools to support the construction of derivations and to encourage the use of advanced logical techniques, there is a separate “Getting Started” document accessed from the “Starting” button. The Derivation module also contains a Frequently Asked Questions document accessed from the FAQ button.

In most cases, the Help document goes on to provide more detail about the module’s facilities.

Help with logic

Four modules contain additional documents to provide help in *learning to do the logic*.

In the Derivation module, the Advice button brings up a document that offers strategic advice on constructing derivations.

In the Symbolization module, the Examples button brings up a document that offers a multitude of examples and explanations of symbolization problems.

In the Truth Tables module, the References button brings up a document that offers the defining tables for the sentential connectives.

In the Recognizing Rules module, the References button brings up a document that discusses the rules of Chapters II and III.

Checking your work

You can check your work at any time by clicking the “Check” button. If your work is correct, the message “Correct” will appear in the upper right hand corner as part of the Problem Line.

If your work contains errors, “Incorrect” will appear as well as one or more error messages. Error messages are sent in different ways in different modules, but generally appear in **red** (or with a **red background**).

Even if you correct the errors, the messages will usually remain until you recheck the work. Error messages are not saved, so you will also need to recheck your work when an erroneous problem (appearing in **red**) that has been saved is recalled.

In the Derivation module a message containing a brief description of the error will appear on each erroneous line. Further explanations can be invoked by clicking the “?” button, which appears along with the message. Not all messages indicate errors, only

messages with a red background indicate **errors**; those with a blue background simply provide **information**.

In the Symbolization module, erroneous choices are marked by an **Error** button, which, when clicked provides further information.

In the Parsing Module, an incorrect choice of connective will cause the symbol to briefly flash **red**. (Correct choices will flash **green**.)

In the Truth Tables module incorrect choices appear in **white** against a red background.

In the Invalidity module, when the model does not invalidate the argument, the truth-value of each premise and the conclusion are indicated in the upper right.

Saving your work

Save your finished or unfinished work on a problem by clicking the “Save” button at the bottom of the Derivation screen. This will save the current version of your work to your “work” folder. If the problem is one that you created, you will be asked to give it a name.

Deleting work

If you wish to start an exercise over again, click the Delete button, and you will be prompted to “Delete the work on this problem?”. If you answer “OK”, the on-screen work, not the saved work, will be deleted. If the exercise is a User Problem, you will also be offered the choice of deleting the problem itself from your “Logic” work-folder.

Printing your work

You can print your work at any time by clicking the Print button at the bottom of the screen. The printout will contain your user information (name, Student ID#, etc.) as a header. You may select as many problems as you wish to print at one time. Use Shift+Click to select a string of adjacent problems, Ctrl+Click to select (or deselect) non-adjacent problems.

In many of the modules, the Print Problems window offers a choice between “Print” and “Print Results”. “Print” prints the complete work for each exercise, one exercise per page. “Print Results” prints all the exercises on a single page, showing only the problem line and “C”, if the work is correct, or “I”, if the work is incorrect or incomplete.

Note that it is the saved version of an exercise that is printed. So always save your work before printing.

You can also print a list of the exercises by first highlighting the problems you wish to print, and then clicking the Print List button.

Submitting your work

You can also submit your work over the Internet to the database of homework submissions established for your course. Click the Submit button at the bottom of the screen. You will be asked to verify course information, and to enter your logic password. You may select more than one exercise to submit. The program will return a message verifying that the submitted work has been received. You can view Assignments and your submissions to them by clicking the “Assignments” button on the Menu.

Closing a Module and/or accessing the Menu

Each module contains a Close button (which closes the module, not the program) and a Menu button (you can Quit the program from the Menu).

Backup & Restore

To back up your work, click “Backup” on the Menu. The database saves your last three backups.

The Restore function will not overwrite existing files, so in order to restore, you must first delete the work in your “work” folder. You can easily delete the files from your current work folder by opening the program, and then clicking “Delete Work/Quit” on the main Menu. The next time you start the program, it will offer the opportunity to *Restore* one of your former backups. You will need your Student ID and your logic password to restore.

Quitting the program

Return to the Menu. If you are in a computer lab, always use “Delete Work/Quit” to quit the program. This will clean up your Work folder so that the next person will not be inconvenienced. If you are not in a place where others expect to use the program, use the Quit button to exit the program. But in any case, always use Backup before you quit.

The Button Bar

In each module there is a button bar at the bottom of the screen. Most buttons are common to all modules.

Button	Function
Select (Select Problem)	Used to select a problem to work on. You may select one of the preset exercises or a User problem that you have already saved.
User (User Problem)	Used to create a “User Problem”, a new problem of your own that is not in the list of exercises.
Rules* (Display Rules)	Pops up a window showing all the inference rules. For derived rules, the theorems necessary to derive the rule are indicated.
Direct** (Direct Entry)	Opens a window into which the symbolization can be entered directly without building a tree.
Check	Initiates an automated check of your work on the current problem, and displays messages describing any errors.

Save	Saves your work on the current problem to the folder LOGIC 2010\work on your local hard disk, where the work will be available to be recalled for editing or printing or submitting.
Delete	Pops up a window asking if you wish to delete the work on the current problem (and begin anew). If the problem is a User problem, you also have the option of deleting the problem itself.
Submit	Submit one or several problems across the Internet to the course database. After clicking the Submit button, you will verify your course, instructor, and TA, and then enter your logic password, and select the exercises you wish to submit, using Ctrl+Click. Click "Submit" at the bottom of the selection window to submit all of the selected problems. After a brief delay, you will receive an acknowledgment that the work has been successfully received. If not, try again.
Print	Pops up the Print window, which lists all of the preset exercises along with all of the User problems that you have saved. The current problem will be highlighted. Use Ctrl+Click to select (or deselect) more than one exercise to print. The Print button at the bottom will then initiate printing of your work, one problem per page. You may also choose the Print Results button, which causes a single page to be printed showing just a list of the selected problems with the results of your work (Correct or Incorrect), but without showing the work itself. You can also print a list of the exercises by first highlighting them and then clicking the Print List button.
Menu	Summon the Menu window.
Close	Close the module window.
Feedback	Send an anonymous (or open) suggestions about any aspect of the course or the program to the instructor, TAs, and program developers. Another source for questions and comments is the Discussion Board on the course website (if there is one). The course website can be accessed from the Menu.
Help	Read a full explanation of how the software for the given module works.
FAQ*	Read the answers to Frequently Asked Questions.
Advice*	Get help with strategies for constructing derivations.
Examples**	Read examples and explanations of symbolization problems.
References***	In the Recognizing Rules module read a discussion of the primitive and derived rules of Chapters II and III. In the Truth Table module, view the defining tables for the sentential connectives.
<i>*Derivations module only.</i>	
<i>**Symbolization module only.</i>	
<i>***At present, Recognizing Rules and Truth Tables module only.</i>	

Common Keystrokes

Several keyboard shortcuts are common to all modules, including **Ctrl+k** (to Check), **Ctrl+s** (to Save), and **Ctrl+p** (to Print).